

DAY-5

Case study practical

1. A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

1. Check the power adapter and outlet

Verify the charger's LED (if it has one) lights up, and test the wall outlet with another device. This checks whether the problem is external power delivery rather than the laptop itself – a dead outlet, faulty cable, or broken adapter brick is a common and easy-to-fix cause.

2. Inspect the charging port and cable connection

Look for bent pins, debris, or a loose-fitting connector at the laptop's charging port, and try wiggling the cable while watching for any charging indicator light. This checks for a physical connection fault, which can interrupt power even if the adapter itself works fine.

3. Try a hard reset (power drain)

Unplug the charger, remove the battery if it's removable, hold the power button for 30 seconds, then reconnect power and try turning it

on (with the battery reinserted if applicable). This checks whether residual static charge or a firmware/power-controller glitch is preventing startup, since a hard reset clears that state.

4 .Test with a known-good charger or battery

If you have access to a compatible charger or a swappable battery, substitute it and attempt to power on. This isolates whether the laptop's own adapter or battery has failed versus a fault in the laptop's internal hardware.

5. Check for signs of life without full boot

Look and listen closely for fan noise, LED lights, coil whine, or screen backlight glow when pressing the power button, even if nothing displays. This checks whether the laptop is receiving power internally but failing later in the boot process (e.g. RAM, motherboard, or display issue) versus a true dead-power state, which points troubleshooting in very different directions.

2. You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Hint: Consider checking both hardware connections and basic display settings.

1. Confirm the CPU is actually running

Check for fan noise, case/motherboard LEDs, and drive activity lights. This confirms the system is powered on and past POST rather than silently hung, which changes whether you're chasing a display issue or a deeper boot issue.

2. Check the monitor's own power and input

Make sure the monitor is powered on (not just in standby – look for a blinking vs solid LED) and that it's set to the correct input source (HDMI, DisplayPort, VGA, etc). Monitors often default back to the wrong input or go to sleep, which looks identical to a 'no signal' hardware fault.

3. Inspect and reseal the video cable

Unplug and firmly reconnect the video cable at both the monitor and the PC end, checking for bent pins or a loose fit. A partially seated or damaged cable is one of the most common causes of a blank screen with an otherwise working system.

4. Test with a different cable, port, or monitor

Swap in a known-good cable, try a different video output port (especially if the system has both integrated and discrete graphics), or connect a different monitor if one is available. This isolates whether the fault is the cable, the port, or the monitor itself.

5. Rule out a GPU or display adapter fault

If using a discrete graphics card, reseal it in its slot and check for any diagnostic beep codes or motherboard error lights. If the system has integrated graphics, try connecting to that port instead. This checks whether the graphics hardware itself is the point of failure.

6. Force a display refresh or safe boot

Try pressing the monitor's auto-detect/source button, or restart the machine and check if it displays BIOS/POST text before the OS loads. This helps determine whether the issue is with the hardware signal path or something that happens later, like a driver or OS-level display failure.

3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

1. Power & pairing

- *Replace/charge batteries; ensure mouse is ON.*
- *Unplug the USB dongle, wait 10s, plug it back in directly (no hub).*
- *Use the dongle's/mouse's Connect/Reset button to re-pair.*
→ *If it works now, the issue was mouse power/pairing.*

2. Test other USB ports

- *Move the dongle to a different USB port (preferably rear, direct on motherboard).*
- *Avoid hubs/extensions; try both USB 2.0 and 3.x if available.*
→ *Works in another port only → original port/path is suspect.*

3. Cross-test on another PC

- *Test the same mouse + dongle (or Bluetooth pairing) on a different computer.*
 → *Fails on other PC too → mouse/dongle likely faulty.*
 → *Works on other PC → problem is with original PC's settings/drivers/ports.*

4. *Check drivers in Device Manager*

- *Device Manager → Mice and other pointing devices / HID.*
- *Uninstall the mouse device, unplug dongle, reboot, plug back in.*
 → *Starts working → driver/software issue on that PC.*

5. *Quick settings check*

- *Settings → Bluetooth & devices → Mouse: ensure it's recognized/enabled; re-pair Bluetooth mice.*
- *Temporarily disable security/macro/remote-desktop tools; test in Safe Mode.*
 → *Works in Safe Mode or after disabling software → software conflict/settings issue.*

Here are three common hardware causes for a gaming-cafe PC that overheats and shuts down, and how I'd check each one.

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each

one.

Hint: Think about fans, dust, and thermal paste.

1. Dust-clogged fans and heatsinks

- *Check: Power off and unplug the system, open the case, and inspect the CPU cooler fins, GPU heatsink, case fans, and PSU intake with a flashlight. Look for thick dust layers blocking airflow.*
- *Test/fix: Use compressed air (holding fans still) to blow dust out of the CPU cooler, GPU, and case vents/filters. Reboot, run a game or stress test, and monitor temps with HWMonitor or similar. A big temperature drop after cleaning confirms dust was the main issue.*

2. Failed or under-spinning cooling fans

- *Check: With the side panel on (or off, safely), power on the system and watch/listen to all fans: CPU fan, GPU fan(s), case intake/exhaust, and PSU fan (if visible). Look for a fan that doesn't spin, spins slowly, or makes grinding/clicking noises.*
- *Test/fix: In BIOS/UEFI, go to Hardware Monitor/Fan Control and verify CPU fan RPM is reported (0 RPM = not spinning or not connected). Reseat fan headers, replace any faulty fans, and retest under load. If the CPU fan was dead or weak, temperatures should normalize once it's fixed.*

3. Dried or poorly applied thermal paste between CPU and cooler

- *Check: If dust is minimal and fans are working but CPU temps are still very high (e.g., >75–80°C at idle, >90–95°C under load), suspect thermal paste.*

- *Test/fix: Power off, remove the CPU cooler, clean old paste from the CPU and cooler base with isopropyl alcohol, apply a small pea-sized amount of new thermal paste, and re-mount the cooler with even pressure. Re-test with monitoring software; a significant temperature drop confirms the paste was the problem.*

If temperatures remain critical after these three checks, I'd then look at airflow layout, ambient room heat, and whether the cooler is adequately sized for the CPU/GPU load in a gaming-cafe environment.